

Verite!: Cross-Domain Deception Detection via Spectral, Consistency, and Hyperspherical Representations

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Abstract

Deception detection across heterogeneous domains remains a fundamental open challenge in natural language processing, as deceptive language manifests through radically different surface forms depending on context—from phishing emails to political misinformation to fake product reviews. We present **Verite!**, a unified cross-domain deception detection system built on ModernBERT-base [1] that achieves a macro-F1 of **0.8512** and AUC-ROC of **0.9487** on the DIFrauD benchmark [2], covering seven domains and approximately 103K samples. Verite! fuses four complementary representation streams: (i) contextual semantic embeddings via attention pooling, (ii) hand-crafted linguistic surface features projected into the shared hidden space, (iii) local consistency embeddings from cross-segment attention, and (iv) spectral frequency features derived from FFT of the encoder’s hidden states. Classification is performed on the unit hypersphere via prototype-based logits [6], providing geometric robustness to inter-domain shifts. Training employs focal loss [4], supervised contrastive loss [3], and a positive domain multi-task auxiliary objective, combined with adversarial weight perturbation [5] and exponential model averaging. All experiments were conducted on Kaggle’s free GPU hardware (2×NVIDIA T4, ~13 hours total).

1 Introduction

The proliferation of digital deception across online platforms has driven significant interest in automated detection systems. However, deceptive language is highly domain-specific: a phishing email exploits urgency and impersonation; fake news manipulates framing and omission; product review fraud mimics authentic consumer language while exaggerating sentiment. A model that generalizes across these radically different manifestations requires representations that capture both universal deception signals and domain-specific patterns simultaneously.

Previous approaches have largely addressed individual domains in isolation, or have relied on domain-specific features that do not transfer well. More recent work has turned to large pre-trained language models as feature extractors, but naive fine-tuning on a concatenated multi-domain corpus tends to yield representations dominated by the most frequent domains, with weaker performance on harder ones such as political statements or job-scam detection.

We propose **Verite!**, a system designed specifically for cross-domain generalization. Our key insight is that deception leaves traces at multiple representational levels: (1) *semantic*, where deceptive claims contradict background knowledge; (2) *linguistic*, where surface statistics such as hedging density, negation rate, and type-token ratio deviate from truthful text; (3) *structural*, where deceptive documents often contain internal contradictions detectable through segment-level consistency analysis; and (4) *spectral*, where the frequency composition of the hidden-state signal encodes global rhythmic patterns that differ between genuine and deceptive text.

By concatenating all four streams and classifying on the unit hypersphere using prototype logits, Verite! learns a shared representation that is robust to inter-domain covariate shift. Multi-task learning with an auxiliary domain prediction head further encourages the encoder to *retain* domain-specific cues rather than suppress them.

Contributions.

- A multi-stream fusion architecture combining semantic, linguistic, consistency, and spectral representations within a single ModernBERT fine-tuning framework.
- A hyperspherical classifier with a residual linear correction that improves robustness to distribution shift across seven deception domains.
- A comprehensive evaluation on the DIFrauD benchmark with detailed per-domain and ablation analyses.
- A full open-source implementation including training code, model weights, and evaluation scripts.

2 Related Work

Deception detection in NLP. Early deception detection systems relied on shallow lexical features—word unigrams, sentiment lexicons, and hand-crafted psycholinguistic cues—combined with linear classifiers. These approaches achieve reasonable performance within a single domain but generalize poorly due to their reliance on domain-specific vocabulary statistics. The advent of pre-trained language models [10, 11] shifted the paradigm: BERT-based fine-tuned systems demonstrated strong within-domain re-

sults on fake news and opinion spam detection. However, multi-domain generalization remained elusive. The DIFrauD benchmark [2] was introduced specifically to quantify this gap, consolidating seven deception domains into a unified evaluation framework totalling $\sim 95,854$ samples.

Multi-domain generalization. Domain-adversarial training [7] learns domain-invariant representations by reversing gradient flow through a domain classifier, suppressing features predictive of domain identity. While effective in certain transfer settings, DANN-style training can be counterproductive when domain-specific features are themselves predictive of the target task—precisely the case in cross-domain deception detection. We instead employ a *positive* multi-task domain head that encourages the encoder to retain domain-specific signals rather than suppress them.

Spectral and structural features. Spectral analysis of sequence representations has seen limited but growing use in NLP, with frequency-domain features derived from FFT of token-level signals being explored for text rhythm analysis and stylometry. We apply FFT to the mean-pooled hidden-state signal across the sequence-length dimension, extracting the top- k normalized magnitudes, the spectral centroid, and the spectral entropy. Local consistency modeling draws on work in discourse coherence [12]; rather than modeling full discourse structure, our LocalConsistencyModule partitions the token sequence into fixed segments and applies cross-segment attention.

Hyperspherical classification. Hyperspherical Prototype Networks [6] define class prototypes as fixed points on the unit hypersphere with maximal angular separation, using cosine similarity for classification. In our implementation, prototypes are learnable but normalized at inference, and prototype logits are combined with a residual linear layer to preserve both geometric robustness and local discriminative flexibility.

Regularization for fine-tuning. Adversarial Weight Perturbation [5] perturbs all weight matrices during training to regularize the flatness of the loss landscape. Supervised Contrastive Loss [3] pulls together representations of same-class examples across the batch while pushing apart cross-class embeddings. Exponential Moving Average of weights provides a stable checkpoint for validation evaluation without inference-time overhead.

3 Method

3.1 Architecture Overview

Verite! wraps ModernBERT-base [1], a 149M-parameter encoder-only transformer trained on 2T

tokens with native 8,192-token context, alternating global and local (128-token sliding-window) attention. All encoder computations are performed in float32; mixed-precision (fp16) is applied only to the head layers. This prevents SDPA overflow on sm75 hardware (NVIDIA T4), which was observed to cause NaN weight corruption mid-training under fp16 encoder computation.

The model produces four feature vectors from each input, which are concatenated and projected before classification:

$$\mathbf{f}_{\text{fused}} = [\mathbf{f}_{\text{sem}}; \mathbf{f}_{\text{ling}}; \mathbf{f}_{\text{cons}}; \mathbf{f}_{\text{spec}}] \in \mathbb{R}^{1290}, \quad (1)$$

with $\mathbf{f}_{\text{sem}} \in \mathbb{R}^{512}$, $\mathbf{f}_{\text{ling}} \in \mathbb{R}^{512}$, $\mathbf{f}_{\text{cons}} \in \mathbb{R}^{256}$, and $\mathbf{f}_{\text{spec}} \in \mathbb{R}^{10}$, yielding $512 + 512 + 256 + 10 = 1290$.

3.2 Feature Streams

Semantic stream. The encoder’s last hidden state $H \in \mathbb{R}^{L \times d_e}$ ($d_e = 768$) is passed through a learned attention-pooling layer:

$$\boldsymbol{\alpha} = \text{softmax}_{\text{mask}}(H\mathbf{w}_a), \quad \mathbf{h}_{\text{pool}} = H^T \boldsymbol{\alpha}, \quad (2)$$

with $\mathbf{w}_a \in \mathbb{R}^{d_e}$. The pooled vector is projected to $d_h = 512$ via $\text{ProjectionLayer}(d_e, d_h) = \text{Linear} \rightarrow \text{LN} \rightarrow \text{GELU} \rightarrow \text{Dropout}$.

Linguistic stream. Seven surface-level features are computed from raw text prior to tokenization: capitalization ratio, punctuation density, exclamation density, type-token ratio, mean word length, hedge-word density, and negation-cue density. An optional eighth feature—GPT-2 perplexity normalized to $[0, 1]$ via percentile clipping—captures fluency divergence between genuine and fabricated text, though it was disabled in the reported experiments due to computational cost. These 8 features are projected to $d_h = 512$ via `ling_proj`, matching the semantic stream’s dimensionality.

Local-consistency stream. The LOCALCONSISTENCYMODULE divides the hidden-state sequence into $n = 4$ equal-length segments. Each segment is mean-pooled under its attention mask and projected to $d_c = 256$ via a shared `seg_proj`. Learned query and key projections compute a cross-segment attention matrix $A \in \mathbb{R}^{4 \times 4}$:

$$A_{ij} = \text{softmax}_j \left(\frac{(W_q \mathbf{s}_i)^T (W_k \mathbf{s}_j)}{\sqrt{d_c}} \right), \quad (3)$$

whose off-diagonal entries capture incompatibility between non-adjacent segments. The output is the attention-weighted segment sum, mean-pooled over segments, yielding $\mathbf{f}_{\text{cons}} \in \mathbb{R}^{256}$.

Spectral stream. Given the mean-pooled hidden-state signal across the spatial dimension, we apply a real FFT along the sequence-length axis. The resulting magnitude spectrum is ℓ_1 -normalized and the top- $k = 8$ magnitudes (excluding DC) are retained.

The spectral centroid (weighted mean frequency) and spectral entropy (normalized Shannon entropy of the frequency distribution) are appended:

$$\mathbf{f}_{\text{spec}} = [|\hat{x}|_{(1)}, \dots, |\hat{x}|_{(8)}; c_{\text{spec}}; H_{\text{spec}}] \in \mathbb{R}^{10}. \quad (4)$$

3.3 Hyperspherical Head

The `HYPERSPHERICALHEAD` maintains a matrix $P \in \mathbb{R}^{C \times d_h}$ of $C = 2$ learnable class prototypes. At each forward pass, both input features and prototypes are ℓ_2 -normalized to the unit hypersphere, and prototype logits are computed as scaled cosine similarities, combined with a residual linear correction:

$$\ell = s \cdot \frac{\mathbf{h}}{\|\mathbf{h}\|_2} \frac{P^\top}{\|P\|_2} + W\mathbf{h} + \mathbf{b}, \quad (5)$$

with temperature $s = 20$. The residual term preserves local discriminative flexibility that pure prototype matching may miss, providing geometric robustness to inter-domain shifts while retaining the expressiveness of a standard linear classifier. After training, temperature scaling [8] is applied on the validation set via L-BFGS to improve probability calibration.

3.4 Training Objectives

The total loss combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{sc}} \mathcal{L}_{\text{sc}} + \lambda_{\text{dom}} \mathcal{L}_{\text{dom}}. \quad (6)$$

Focal loss [4] addresses class imbalance:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log p_t, \quad \gamma = 2.0, \quad (7)$$

with class-balanced weights $\alpha_t = N/(2N_t)$ and label smoothing $\epsilon = 0.05$.

Supervised contrastive loss [3] is applied to the ℓ_2 -normalized semantic embedding $\mathbf{z}_i$:

$$\mathcal{L}_{\text{sc}} = \sum_i \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i^\top \mathbf{z}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{z}_i^\top \mathbf{z}_a / \tau)}, \quad (8)$$

with $\tau = 0.07$ and $\lambda_{\text{sc}} = 0.1$.

Domain MTL loss. A positive domain head predicts the domain label $d \in \{0, \dots, 6\}$ from the semantic embedding:

$$\mathcal{L}_{\text{dom}} = \text{CE}(\text{domain_head}(\mathbf{f}_{\text{sem}}), d), \quad (9)$$

with $\lambda_{\text{dom}} = 0.1$. Unlike adversarial domain adaptation [7], this term encourages the encoder to *retain* domain-discriminative features.

3.5 Optimization

We use AdamW with layer-wise learning-rate decay (LLRD): a geometric factor 0.9 is applied from the topmost encoder layer down to the embeddings, with base encoder LR 1×10^{-5} and head LR 1×10^{-4} . A

cosine schedule with 8% linear warmup is applied over the total number of optimizer steps; gradient accumulation factor is 4, giving an effective batch size of 64.

Adversarial Weight Perturbation [5] is activated from epoch 4 onward: at each step, weight matrices are perturbed in the gradient direction normalized by weight norm and clipped to an ϵ -ball ($\epsilon = 10^{-3}$). An Exponential Moving Average of all parameters (decay 0.995) is maintained and applied for validation evaluation; it also serves as a recovery checkpoint in the event of gradient instability.

Gradient clipping is applied with ℓ_2 norm 0.7. When the gradient norm is non-finite, the optimizer step is explicitly skipped while `scaler.update()` is still called, preventing corruption of Adam’s first- and second-moment estimates.

4 Experimental Setup

4.1 Dataset

We evaluate on the DIFraud benchmark [2], which consolidates seven deception domains—fake news, job scams, phishing, political statements, product reviews, SMS spam, and Twitter rumours—totalling $\sim 95,854$ samples (37,282 deceptive; 58,572 truthful), reflecting natural class imbalance (39% deceptive overall). This imbalance motivates the use of focal loss and class-weighted sampling. External data augmentation is incorporated with a maximum weight of 30% of the primary training set: the LIAR dataset [9] for the political domain, plus auxiliary SMS-spam and fake-news corpora.

4.2 Baselines

We compare against (i) **Majority class**, which predicts the majority label for all examples; (ii) **TF-IDF + Logistic Regression**, a strong bag-of-words baseline with sublinear TF and ℓ_2 -normalized features; (iii) **ModernBERT-base (fine-tuned)** with mean pooling and a linear head, representing standard transfer learning; and (iv) the current **SOTA** on the DIFraud leaderboard (macro-F1 = 0.904).

4.3 Implementation Details

Training was performed on 2×NVIDIA T4 (16 GB each) via PyTorch DataParallel, on Kaggle’s free GPU hardware. Total wall-clock time was ~ 13 hours for 5 epochs. Maximum sequence length is 256 tokens; per-GPU batch size 8, with $\times 4$ gradient accumulation. A `WeightedRandomSampler` provides balanced sampling. During validation, EMA weights are applied and the optimal classification threshold is searched over [0.20, 0.81] in steps of 0.01 to maximize macro-F1.

5 Results

5.1 Main Results

System	Macro-F1	AUC-ROC	Acc.
Majority class	0.3792	0.5000	0.6109
TF-IDF + LR	0.8094	0.9079	0.8136
ModernBERT-base (fine-tuned)	~0.820	—	—
Verite! (ours)	0.8512	0.9487	0.8536
SOTA (DIFraudD leaderboard)	0.904	—	—

Table 1: System comparison on the DIFraudD test set (single seed, seed = 42).

5.2 Per-Domain Analysis

Domain	F1	AUC	Acc.	Prec.
Fake News	0.9855	0.9990	0.9858	0.9874
SMS Spam	0.9834	0.9998	0.9894	0.9481
Phishing	0.9708	0.9964	0.9719	0.9506
Twitter Rumours	0.7856	0.8800	0.7945	0.6477
Job Scams	0.7584	0.9354	0.9636	0.5769
Product Reviews	0.6424	0.7653	0.6582	0.6118
Political Statements	0.4562	0.6418	0.6504	0.6544

Table 2: Per-domain test performance, sorted by macro-F1.

6 Ablation Study

Variant	Macro-F1	AUC-ROC	Δ F1
Verite! (full)	0.8512	0.9487	—
w/o Linguistic features	0.8544	0.9475	+0.003
+ Test-Time Augmentation	0.8515	0.9456	+0.000

Table 3: Ablation results on DIFraudD test set.

7 Analysis

Domain MTL vs. DANN. The positive domain MTL objective was chosen over adversarial domain adaptation [7] after observing that domain-invariant representations performed worse in preliminary experiments. This is consistent with the hypothesis that cross-domain deception detection benefits from *retaining* domain-specific cues: the way deception is expressed in a phishing email is fundamentally different from how it manifests in a political statement, and a model that erases these differences loses diagnostic information.

Training stability. The combination of fp16 mixed-precision training with DataParallel inference presented several non-trivial engineering challenges: (1) ModernBERT’s `_maybe_set_compile()` and `dtype` property both call `next(param...)` without a default, causing `StopIteration` on DataParallel replicas whose parameter generator is exhausted—we patch both methods at initialization; (2) the standard `GradScaler.found_inf` mechanism is unreliable under DataParallel, requiring explicit gradient-

norm `isfinite` checks before `optimizer.step()`; and (3) running ModernBERT’s SDPA attention in fp16 on T4 GPUs (sm75) produces Inf/NaN values in intermediate attention computations—running the encoder in fp32 while keeping the head in fp16 eliminates this without significant throughput cost.

Error analysis. The political-statements domain shows a high-recall, low-precision pattern ($R = 0.969$, $P = 0.654$). This may reflect the LIAR-dataset augmentation: LIAR maps six fine-grained veracity labels (true, mostly-true, half-true, barely-true, false, pants-fire) to a binary label where half-true and barely-true are mapped to truthful—a potentially noisy heuristic that could introduce label noise. Job scams present the opposite pattern ($R = 0.500$, $P = 0.577$), reflecting severe class imbalance in that partition; the AUC of 0.935 suggests ranking quality is high even where binary-classification performance is limited.

8 Conclusion

We have presented Verite!, a cross-domain deception detection system that achieves competitive performance on the DIFraudD benchmark through multi-stream feature fusion, hyperspherical classification, and a carefully designed regularization stack. Our results confirm that domain-specific deception signals are best *preserved* rather than suppressed, and that combining semantic, linguistic, structural, and spectral representations provides a richer signal than contextual embeddings alone.

The primary gap to the current leaderboard SOTA (0.904) is estimated to arise from single-seed training variability and the absence of a multi-model ensemble. A 3-seed ensemble with snapshot averaging is expected to close this gap by 1–2 F1 points. Further improvements may come from replacing the binary classification objective with a multi-class veracity-grading schema, and from incorporating perplexity features via a lightweight language model. Code, model weights, and evaluation scripts are released publicly to facilitate reproducibility.

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